SUPPORTING ADULT GUIDANCE

Can I write the line?

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner combines the two numbers they have studied separately — the y**-intercept c** and the **gradient m** — into the equation of a line, y **= mx + c**. The emphasis is on **reading c and m from a line or a situation**, writing them in the right places, and keeping their **signs**.

Driving Question: *How do the gradient and the intercept become an equation?*

The mathematics, in plain terms

A straight line is completely fixed by just two numbers. The **gradient m** is how steep it is — how much y changes for every 1 across — and its sign says which way the line leans (up for positive, down for negative). The y**-intercept c** is where the line crosses the y-axis: the value of y when $x = 0$. Put them together and you have the line's equation: y **= mx + c**. For a line that crosses at (0, 4) and climbs 3 for every 1 across, m = 3 and c = 4, so the equation is y **= 3**x **+ 4**. The order and the places matter: the gradient **multiplies** x, and the intercept is **added on its own**. Two slips are worth naming out loud. The first is **swapping m and c** — writing $y = 4x + 3$ for the line above. The second is **losing a sign** — a falling line has a **negative** gradient ($y = -2x + 5$), and a line can cross below the origin, giving a **negative** intercept ($y = x - 3$). The real world is full of these: a taxi with a £2 flag fee then £1 per kilometre is y **=** x **+ 2**; a phone with £10 credit falling £2 a day is y **=** $-$**2**x **+ 10**. The fixed part is c, the rate is m. Being able to write the line is what makes the next lessons possible — comparing two lines, finding where they meet, and using them as models.

Two numbers, two places

The heart of this lesson is simple to say and worth protecting: a line needs exactly two numbers, and each has its place in $y = mx + c$. The gradient m goes in front of x; the intercept c is added on its own. Keeping those two roles distinct is what prevents the classic swap. It helps to build the equation from the two things the learner can already find — the crossing point gives c, the 1-across step gives m — so the equation feels assembled rather than conjured. Signs deserve deliberate attention: a falling line means a negative m, and a line crossing below the origin means a negative c. Grounding the equation in a real fixed-fee-plus-rate situation keeps m and c meaningful — the fee is c, the rate is m — and sets up the comparing and intersecting of lines that the unit turns to next.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Line Detective: read c (the crossing) and m (the step) from a line, then write $y = mx + c$.	~12 min

Movement	Equation Builder: set the c and m sliders to match a target line, and watch the equation build.	~12 min
Reflection	Right or slip?: is the written equation right, or are m and c swapped or a sign lost?	~10 min
Creativity	Write a real line: turn a starting value and a steady rate into y = mx + c, signs included.	~6 min
Rest	A pause after bringing two separate ideas together into one equation.	~3 min
Nutrition	Contextual (Mode A): fixed fee plus a rate — taxis, gyms, tariffs — is every y = mx + c.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Swapping m and c (writing the intercept in front of x)

Repair: “Which number multiplies x — the gradient or the intercept? Put the gradient in front of x, and add the intercept on its own.”

Dropping the minus sign on a falling line

Repair: “As x increases, does y go up or down? Down means m is negative — keep the minus.”

Missing a negative intercept

Repair: “Where does the line cross the y-axis? If it is below zero, c is negative.”

Reading the gradient off a single point instead of the step

Repair: “Go 1 across and count the up or down — that is m. A single coordinate is not the gradient.”

Thinking one number is enough to write the line

Repair: “Could two different lines share this gradient? Or this intercept? You need both to pin one line down.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading the y-intercept from a graph, finding a gradient (the 1-across step), and knowing which number in $y = mx + c$ is which. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the two numbers and their places with the learner:

- “Where does the line cross the y-axis? That is c.”
- “Go 1 across — how far up or down? That is m (keep the sign).”
- “Now write y = mx + c — which number goes in front of x, and which is added on?”
- “In this real situation, what is the fixed part (c) and what is the rate (m)?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Does it matter which way round m and c go?”

Yes. The gradient m multiplies x, and the intercept c is added on its own. Swapping them describes a different line — a quick check is to read the crossing point off the equation (the number without an x) and see if it matches the graph.

“What if the line falls, or crosses below zero?”

Then a sign is negative. A falling line has a negative gradient (m < 0); a line crossing below the origin has a negative intercept (c < 0). Writing the sign in is part of writing the line correctly.

“How does this set up the next lessons?”

Once a learner can write a line, they can write two — and then compare them: parallel lines that never meet, and intersections where two lines are equal (a break-even or equal-value point). Writing the equation is the key that unlocks the rest of the unit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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